DAY 23 | PRE-READ

Searching by Meaning

A light read before Day 23. About 15 minutes, no code required.

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Why read this?

Yesterday you saw that an AI assistant needs the right passages from your documents. Tomorrow you learn how computers find passages by meaning, not just by matching words, and how to make that search fast and accurate.

1. Meaning as a location

Imagine a giant map where every sentence has a location, and sentences with similar meanings are placed close together. "How many holidays do I get?" sits right next to "Employees receive 18 days of paid leave", even though they share almost no words. The location of a sentence on this map is its **embedding**: a list of numbers that captures meaning.

2. A library for these locations

A **vector database** stores millions of these locations and finds the nearest ones to a question in milliseconds. Like a librarian who knows the shelves well, it does not check every book; it goes straight to the right area. This shortcut is called **approximate nearest neighbour** search.

3. Two kinds of search, better together

Search type	Strength	Weakness
Keyword search	Exact terms like error code VPN-417 or a product name	Misses different wording for the same idea
Meaning search	Understands paraphrases and synonyms	Can miss rare exact codes

Combining both is called **hybrid search**, and it is what most serious systems use.

4. A second opinion

Search returns a shortlist. A **re-ranker** then reads each candidate carefully against the question and reorders them, like a hiring manager reviewing the shortlist a recruiter sent. The best evidence goes to the model, which answers with citations, or admits it does not know.

Key idea: good answers come from good evidence. Most RAG problems are search problems.

5. Think about these before class

- Think of a time a search engine misunderstood you. Was it a keyword or a meaning problem?
- Which exact codes or names in your work would keyword search handle better?
- Why might an assistant that says "I don't know" be more trustworthy?

Imi (意味): meaning